3AL Hand-in 4 2024

Due: Friday 3 May 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

   (a) If G is a finite group and $p \mid |G|$, then all elements of order p are contained in a Sylow p-subgroup.

   (b) If H is a Sylow p-subgroup of a group with $|H| = p^n$, then all elements with order p^n are contained in H.

   (c) The number of conjugates of a Sylow p-subgroup is a multiple of p.

   (d) If G has an abelian Sylow p-subgroup, then all Sylow p-subgroups of G are abelian.

   [4 marks]

2. Use the Sylow Theorems to prove that if G is a group of order 33, then $G \cong \mathbb{Z}_3 \times \mathbb{Z}_{11}$.

   [8 marks]

Maximum mark = 10

Available marks = 12